# Working memory updating as a predictor of academic attainment

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There is growing evidence supporting the importance of executive functions, and specifically working memory updating, for children's academic achievement. This study aimed to assess the specific contribution of updating to the prediction of academic performance. Two updating tasks, which included different updating components, were administered to 97 fourth-grade children. The keeping track task involves retrieval and substitution of information, while the numerical updating task also includes a transformation component. Academic attainment was assessed through standardized tests of verbal comprehension, arithmetic operations, mathematical problems, and an assessment made by the teacher. The relative contribution to academic attainment of the updating measures and measures related to intelligence. Results showed that both updating tasks are predictive measures of academic attainment, although the numerical updating task appeared to be a more consistent predictor of children's performance. The relationship between updating and academic attainment is discussed, and possible educational implications are considered. The role of the transformation component of working memory updating is highlighted. This component could make a distinct and independent contribution to performance and, by extension, could be particularly relevant to the prediction of academic achievement.

**Keywords:** working memory; working memory updating; central executive; transformation; academic attainment.

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## Introduction

Working memory (WM) is the mechanism responsible for the simultaneous storage and processing of information when engaging in complex cognitive activities (Miyake & Shah, 1999). Working Memory Updating (WMU) is a central process in mental architecture which reflects the prototypical dynamic operation of WM (e.g., Friedman, Miyake, Young, DeFries, Corley, & Hewitt, 2008; Schmiedek, Hildebrandt, Lövdén, Wilhelm, & Lindenberger, 2009). The updating function requires the monitoring of incoming information for relevance to the task, and then appropriately revising the items held in WM by replacing older, outdated information with newer, more relevant information (e.g., Morris & Jones, 1990). In recent years, there has been some theoretical controversy concerning whether complex, traditional WM and WMU measures should be considered interchangeable given that several studies have reported only weak correlations between them (Kane, Conway, Miura, & Colflesh, 2007), while others have shown strong correlations (Schmiedek et al., 2009; Wilhelm, Hildebrandt, & Oberauer, 2013).

It could be argued that complex WM and WMU tasks may require different processes. According to the previous definition, complex WM tasks would entail the simultaneous storage and processing of information. For instance, in the well-known listening span task (Daneman & Carpenter, 1980), participants have to simultaneously answer compression questions and recall the final word of each sentence. Therefore, this involves combining two tasks which are not entirely related. Moreover, once the final word has been stored, that information has no longer to be substituted. Conversely, updating tasks, such as the n-back (Kirchner, 1958), require continuous updating of the information maintained in memory and do not necessarily involve dual tasking or the concurrent processing of information other than that which is to be memorized. This study focused on updating tasks characterized by continuous substitution of the information to be recalled, rather than on complex WM tasks.

WMU has been identified as a cognitive factor which accounts for a significant proportion of individual differences in executive functioning (e.g., Miyake, Friedman, Emerson, Witzki, Howerter, & Wager, 2000), and some studies have revealed the existence of developmental changes in the effectiveness with which WM memory is updated (e.g., Garon, Bryson, & Smith, 2008; Huizinga, Dolan, & Van der Molen, 2006; Lechuga, Moreno, Pelegrina, Gómez-Ariza, & Bajo, 2006; Pelegrina, Borella, Carretti, & Lechuga, 2012).

It is therefore not surprising that a growing number of studies demonstrate the important role of WMU in accounting for individual differences in attainment in different academic areas, such as reading comprehension, mathematical problems or arithmetical tasks. Several researchers have shown that students with low reading skills have problems updating the content of their memory appropriately (e.g., Borella, Carretti, & Pelegrina, 2010; Carretti, Cornoldi, De Beni, & Romano, 2005; Cornoldi, Drusi, Tencati, Giofrè, & Mirandola, 2012; Palladino, Cornoldi, De Beni, & Pazzaglia, 2001, although see Swanson, Howard, & Saez, 2006). The results of a meta-analysis carried out by Carretti and collaborators (Carretti, Borella, Cornoldi, & De Beni, 2009), on 18 studies in which reading comprehension performance was analyzed, confirm that updating ability is lower in individuals with comprehension difficulties.

WMU is also involved in performing arithmetical operations and mathematical problems. The studies carried out to investigate the role of WMU in mathematics also appear to support the idea that updating ability is an important factor in mathematical performance. It has been shown that children with low mathematical skills perform worse in WMU tasks (e.g., Passolunghi & Pazzaglia, 2004, 2005; Swanson & Beebe-Frankenberger, 2004). Recently, Van der Ven, Kroesbergen, Boom, and Leseman (2012) have found a strong relationship between WMU and the mathematical learning process. In fact, updating ability has been shown to be the best predictor of individual differences in numerical magnitude skills, even in five-year old children (Kolkman, Hoijtink, Kroesbergen, & Leseman, 2013).

Performance in WMU tasks has also been used to predict general academic attainment in typically developing children. For example, St Clair-Thompson and Gathercole (2006) administered a series of executive WM tasks, including shifting, updating, inhibition, and span WM, to a group of 11-year-old children in order to predict their academic attainment in reading and writing, mathematics and science. They reported a consistent correlation between performance in WMU tasks and academic attainment, leading the authors to conclude that WMU is a domain-general skill crucial in academic tasks. Using similar logic, Van der Sluis, de Jong, and Van der Leij (2007) also examined the relationships between these executive functions and, although correlations were weaker than those obtained by St Clair-Thompson and Gathercole (2006), they also found that WMU contributes to explaining the variance in performance of reading and arithmetic tasks, and in verbal and non-verbal reasoning tests.

Recently, some authors have proposed that WMU may not be a single unitary process, but may rather entail different subprocesses. Specifically, Ecker, Lewandowsky, Oberauer, and Chee (2010) have demonstrated that WMU can be broken down into different components which make independent contributions to updating performance. The three components identified were  retrieval, transformation, and substitution. Thus, most of the WMU tasks involve accessing or retrieving some memory contents that were previously stored in WM (e.g., Oberauer, 2003). In addition, the task may or not require the manipulation and transformation of stored information (e.g., Carretti, Cornoldi, & Pelegrina, 2007; Ecker et al., 2010). Finally, WMU involves the substitution of the no-longer relevant information with new content (e.g., Kessler & Meiran, 2008; Lendínez, Pelegrina, & Lechuga, 2011, 2014). According to Ecker and colleagues (2010), different WMU tasks vary in the extent of involvement of these processes, which may give rise to differences in their predictive capacity. In addition, while transformation and access are related to WM, the substitution component appears to be independent. Of these components, transformation has the greatest impact on updating accuracy and speed.

Previously reviewed studies on the role of WMU in academic skills have included WMU tasks that mainly tap the substitution and retrieval components. For instance, tasks such as keep track, letter memory, running task, or n-back require participants to remember the last presented items, according to certain criteria. In addition, participants have to replace the previously memorized information with other, newer information as the task progresses. Thus, these tasks require both the retrieval and substitution of information, although they do not include the transformation component. In fact, there is usually no relationship between the old and new information.

The transformation component could however be highly relevant to many different educational activities. This is evident in certain mathematical problems, or in arithmetic operations that have to be solved mentally and involve completing intermediate steps before a solution is reached. The transformation of initially stored information is also required in order to comprehend a text. This occurs when a description involves modification of a previous situation. For instance, when a character moves to a different location certain details might have to be changed, depending on the initial situation. Thus, a transformation of the initial representation must be made in order to update the information necessary to understand the text.

The objective of the present study was to explore the contribution of updating to the prediction of general academic attainment, in different areas, in typically developing children. Given that the updating tasks used in the previously described studies did not include the transformation component, in order to better determine the role of updating in academic attainment two WMU tasks were chosen, one with and one without the transformation component.

A numerical updating task (Oberauer & Kliegl, 2001) was selected which, to our knowledge, has not previously been used in this context. In this task, participants are first shown a series of numbers, each in a separate frame, followed by a series of simple arithmetical operations which appear consecutively in individual frames. Participants have to apply each operation to the number in the corresponding frame and memorize the result for that frame. Therefore, participants have to recall the content of each frame, carry out the arithmetical calculation and replace the obsolete number with the result of the operation. The numerical updating task involves, in addition to retrieval and substitution components, transformation of the stored information. According to Ecker et al. (2010), the latter component has the strongest impact on WMU performance. In their study, conditions that required transformation led to lower performance than conditions without transformation requirements.

We added the keep-track task, which has been used previously in a number of studies (e.g., St Clair-Thompson & Gathercole, 2006; Van der Sluis et al., 2007). In this task a series of words belonging to different categories (for example, tools, animals and parts of the body) are presented, and participants have to recall the last element shown for each category (for example, hammer, dog and leg). Participants have to memorize the last presented word for each category while discarding any previous items associated with that same category. The keep-track task, which requires retrieval and substitution of information, but not transformation, has been found to predict performance in different measures related to academic achievement to varying extents.

In order to better determine the unique contribution of WMU to academic outcomes, some standardized measures of vocabulary and non-verbal reasoning were included, given that both crystallized and fluid intelligence have been shown to be powerful predictors of academic attainment (e.g., Deary, Strand, Smith, & Fernandes, 2007). Crystallized intelligence refers to mental operations which strongly rely on previous knowledge, while fluid intelligence is related to operations not requiring reference to preexisting knowledge (e.g., Cornoldi, 2006).

WMU is the executive function which best predicts performance on cognitive tests of fluid and crystallized intelligence (Friedman, Miyake, Corley, Young, DeFries, & Hewitt, 2006). In particular, WMU has been shown to be a critical factor in predicting performance on crystallized (Brydges, Reid, Fox, & Anderson, 2012) and fluid intelligence in children (e.g., Belacchi, Carretti, & Cornoldi, 2010; Brydges, et al., 2012; Duan, Wei, Wang, & Shi, 2010). Given that the relationship between these constructs is not perfect, they may differentially contribute to academic attainment. A comparison between WMU and verbal and non-verbal intelligence-related measures will allow us to assess whether updating makes any additional contribution towards explanation of academic attainment.

In summary, we aimed to explore the contribution of two updating memory tasks, the keep-track task and the numerical updating task, to the prediction of academic achievement. Based on previous research, it was expected that both tasks would predict performance in the different achievement measures. In addition, and given that the numerical updating task requires an additional updating component, namely the transformation of stored information, it might be expected that the numerical task would have greater predictive effectiveness than would the keep-track task.

## Method

### *Participants*

The participants in this study were 97 children (45 boys and 52 girls) drawn from the 4$^{th}$ grade of primary (elementary) education. Children had an average age of 9.58 years (*SD* = .35). All of the participants were from middle-class families, with similar socio-cultural characteristics. None of the participants had special educational needs. Participant and parental consent was obtained.

### *Measures*

#### *Academic attainment measures*

The ability to solve numerical-verbal problems, simple arithmetical operations, and verbal comprehension was measured using the corresponding subtests from the BADyG E2 battery (Yuste, 1998).

*Mathematical problems*. The test consisted of 24 arithmetic word problems which could be solved by applying a simple arithmetical operation (addition, subtraction or multiplication) to the numbers given (e.g., "I have 6 pencils, I'm given 5 colored pencils, how many do I have now?). The time available was 10 minutes. The final score reflected the number of correct answers. The reliability coefficient reported in the manual for 9-year-old children was .91.

*Arithmetic operations*. This test comprised 24 pairs of addition and subtraction sums (e.g., 6+1 and 8-2) with one- or two-digit numbers. The objective was to determine which sum was greater. The time available was 8 minutes. The final score corresponded to the number of correct responses. The value of Cronbach's α reported in the manual was .85.

*Verbal Comprehension*. This task involved responding to a series of 24 questions after listening to a text read aloud by the examiner. The time available was 8 minutes. The final score corresponded to the number of correct responses. The split-half reliability reported in the manual was .79.

*Teacher Assessment.* This measure was based on the overall academic achievement of the child, as assessed by the teacher, and rated on a scale of 0 to 10.

*Intelligence measures*

*Reasoning.* Non-verbal reasoning was assessed using the corresponding subtest from the "SRA Test of Educational Ability", TEA-1 (Thurstone & Thurstone, 1974). This test assesses capacity for abstract reasoning without reference to preexisting knowledge. The test comprised 30 series of four drawings. In each series students had to choose which of the four drawings was different from the others, in accordance with a visual criterion. Of the four geometric figures/drawings, three had common features (e.g., straight lines) but the forth included a unique characteristic (e.g. a curved line). The final score corresponded to the number of correct answers. The value of Cronbach's α reported in the manual was .72.

*Vocabulary*. In order to obtain a verbal measure related to intelligence, the vocabulary subtest from the Primary Mental Abilities, PMA test (Thurstone & Thurstone, 1976) was administered. This measure makes it possible to assess the contribution of knowledge-based skills to academic attainment. The test included 50 items involving a word and four possible synonyms. Participants had to select, from among the four synonyms, the one which had the same meaning as the original word.

The time limit was 4 minutes. The final score corresponded to the number of correct responses. The split-half reliability reported in the manual was .91.

*WMU tasks*

Two tasks were used to assess updating ability, one with linguistic (keep-track task), and the other with numerical (numerical updating task), content.

*Keep-track task*. This task was adapted from Salthouse, Atkinson and Berish (2003). In order to make the task easier for children, the memory load and number of updates required for each trial was reduced. Before beginning participants were shown three or five categories (depending on the load). Subsequently, they were shown a variable sequence of words belonging to the categories (from 4 to 8). Finally, participants had to recall the final word which belonged to each category. The “Spanish Category Norms for Children” developed by Goicoetxea (2000) was used to construct the lists. The categories included were: trees, musical instruments, furniture, colors and parts of a house. The task consisted of 18 experimental and 2 practice trials. In order to prevent participants knowing the number of elements shown in each category, the lists were of different lengths (from 7 to 13 words). Each trial began by showing in the bottom third of the screen the names of the categories to which the words to be studied belonged. The list of categories was visible throughout the test. Following an audible signal, words were sequentially presented in the center of the screen at a rate of one word every two seconds. Once all the words had been shown, participants were asked to write down the final word of each category. The percentage of correctly-recalled items was taken as the dependent measure. The split-half reliability estimate for the present study was .68.

*Numerical updating task*. This task was adapted from Oberauer and Kliegl (2001), such that presentation time and memory load were more suitable for children.

The task consisted of a series of simple arithmetical operations (addition and subtraction) using numbers shown in frames in different places on the screen. After displaying 2 or 3 initial numbers (depending on the load) for 4000 ms, the numbers disappeared. Subsequently, a variable number of simple calculations (from 2 to 5) was displayed in sequence (e.g. “+2”, “-4”) in the different frames, for 2500 ms each. Participants had to apply each operation to the number in the frame, calculate the sum and maintain the new number associated with the frame for future recall. The sum was never lower than 1 nor higher than 9. Following the various sequential operations, a question mark appeared in each of the squares which had initially contained numbers (see Figure 1). This signaled to the students that they had to write down the final value of each square. The task comprised 42 tests, of which 2 were trials and the remaining 40 experimental tests. The percentage of correctly-recalled numbers was considered as the dependent measure. The estimated value of Cronbach’s α for the present was .85.

(Insert Figure 1 here)

For the programming and presentation of the two updating tasks, E-prime (Schneider, Eschman, & Zuccolotto, 2002) was used. A portable computer was used for presentation of stimuli.

***Procedure***

The tasks were administered in two separate sessions by the first and third authors. In the first session, the three standardized academic attainment tests, and the two standardized intelligence tests, were collectively administered in the classroom. These tests took approximately 60 minutes, with a break around the half-way point. In the second session, the two WMU tasks were administered to small groups of 5-6 children. This session took place in a quiet room at their school and lasted approximately 60 minutes. The two sessions took place on separate days but were relatively close to each other (within a period of one week).

**Results**

As a first step, Pearson correlations among the school performance measures and non-verbal reasoning, vocabulary, keep-track and numerical updating tasks were calculated. The results of these analyses[1], including means and standard deviations, are presented in Table 1. All of the correlations were statistically significant ($p$ < .001 bilateral).

(Insert Table 1 about here)

A series of regression analyses were carried out in order to predict each of the measures of academic attainment: mathematical problem-solving, arithmetical operation completion, verbal comprehension and teacher assessment. To examine whether the four predictor variables contributed to the prediction of each of the academic attainment measures, all variables were entered simultaneously into the regression analysis. Squared partial correlations represent the unique contribution of each variable.

An additional goal was to identify the unique variance accounted for by updating and intelligence measures when predicting academic attainment. To this end, a series of two-step hierarchical multiple regressions were performed.

As can be seen in Table 2, for the mathematical problems up to 50% of the variance in scores was accounted for by performance in updating and intelligence measures, $R^2 = .50$, $F$ (4, 92) = 38.52, $p < .001$, and when all variables were entered simultaneously in the regression analysis. In order to identify the unique variance accounted for by updating versus intelligence measures, two-step hierarchical multiple regressions were conducted in which the updating variables were entered at the first step, followed by the intelligence variables (model 1). In this case, updating accounted for 45% of the variance in mathematical problem-solving performance, $\Delta R^2 = .45$,

$F_{change}$ (2, 94) = 38.52, $p < .001$, while intelligence explained an additional 5% of the variance, $\Delta R^2 = .05$, $F_{change}$ (2, 92) = 4.71, $p = .011$. However, when the order of inclusion was reversed, and the intelligence variables were entered at the first step (model 2), intelligence accounted for 37% of the variance, $\Delta R^2 = .37$, $F_{change}$ (2, 94) = 27.26, $p < .001$, while updating measures accounted for an additional 13%, $\Delta R^2 = .13$, $F_{change}$ (2, 92) = 12.40, $p < .001$. Therefore, although both variables made a unique contribution to the prediction of mathematical problem-solving performance, it was updating which accounted for a greater proportion of the additional variance in performance. The squared partial correlations (see Table 2) showed that the largest independent and unique contribution was for the numerical updating task, which represented a unique contribution of 13% of the variance, whereas the reasoning test accounted for 8% and the keep-track task accounted for an additional 4%. Therefore, the regression analyses, in combination with the partial correlations, showed that the most relevant independent predictors in mathematical problem-solving performance were the updating variables, specifically the numerical updating measure.

(Insert Table 2 about here)

For arithmetical operations, updating and intelligence measures together explained up to 37% of the variance in scores, $R^2 = .37$, $F$ (4, 92) = 13.65, $p < .001$. As in the previous analyses, in order to determine the relative contribution of updating versus intelligence measures the updating variables were entered at the first step, followed by the intelligence variables, in the two-step hierarchical multiple regressions. In model 1, 32% of the variance was accounted for by updating, $\Delta R^2 = .32$, $F_{change}$ (2, 94) = 22.42, $p< .001$, while intelligence explained an additional 5% $\Delta R^2 = .05$, $F_{change}$ (2, 92) = 3.63, $p = .030$. However, when the order of inclusion was reversed (model 2), intelligence accounted for 23% of the variance, $\Delta \mathrm{R}^2 = .23$, $F_{change}$ (2, 94) = 13.89, $p<$

.001, while updating added up to 14%, $\Delta R^2 = .14$, $F_{change}$ (2, 92) = 10.58, $p < .001$. Therefore, updating accounted for a greater amount of variance in arithmetic operations. The largest independent and unique contribution was for the keep-track variable, which represented a unique contribution of 10% of the variance (see Table 2). The vocabulary variable accounted for an additional 7% of the variance, with the numerical updating variable accounting only for an additional 5%, even when that task involved arithmetical operations. The regression analyses, in combination with the partial correlations, showed that, for the arithmetical operations, the most relevant independent predictors were the updating variables, in this case the keep-track updating measure.

The third set of analyses examined the relative contribution of updating and intelligence measures to the prediction of verbal comprehension. Up to 35% of the variance in performance was predicted by the updating and intelligence measures, when all variables were entered simultaneously into the regression analysis, $R^2 = .35$, $F$ (4, 92) = 12.31, $p < .001$. When the updating variables were entered at the first step in the two-step hierarchical multiple regressions (model 1), up to 22% of the variance was accounted for by updating, $\Delta R^2 = .22$, $F_{change}$ (2, 94) = 12.95, $p < .001$, while intelligence explained an additional 13% of variance, $\Delta R^2 = .13$, $F_{change}$ (2, 92) = 9.36, $p < .001$. However, when the order of inclusion was reversed (model 2), intelligence accounted for 31% of the variance, $\Delta R^2 = .31$, $F_{change}$ (2, 94) = 20.82, $p < .001$, while updating only added a small but significant increase of 4%, $\Delta R^2 = .04$, $F_{change}$ (2, 92) = 2.93, $p = .058$. Therefore, for the verbal comprehension measure, intelligence variables accounted for a greater percentage of the variance. The largest independent and unique contribution was for the vocabulary measure, which represented a unique contribution of 16% of the variance, whereas numerical updating contributed only an additional 4%

(see Table 2). This result demonstrates that knowledge-based skills are a determinant of verbal comprehension.

The final set of analyses compared the relative contribution of updating and intelligence to the prediction of teacher assessment. Performance in updating and intelligence measures together accounted for 49% of the variance in scores, $R^2 = .49$, $F(4, 92) = 22.22$, $p < .001$. The relative contribution of updating versus intelligence measures, to the prediction of teacher assessment, was evaluated again via the two-step hierarchical multiple regressions analyses. In model 1, updating accounted for 46% of the variance, $\Delta R^2 = .46$, $F_{\text{change}}(2, 94) = 39.76$, $p < .001$, while intelligence added a small yet significant additional 3% of variance, $\Delta R^2 = .03$, $F_{\text{change}}(2, 92) = 3.00$, $p = .055$. However, when the order of inclusion was reversed (model 2), intelligence accounted for 33% of the variance, $\Delta R^2 = .33$, $F_{\text{change}}(2, 94) = 23.25$, $p < .001$, while updating explained an additional 16% of variance, $\Delta R^2 = .16$, $F_{\text{change}}(2, 92) = 14.51$, $p < .001$. Therefore, updating accounted for a greater amount of unique variance in teacher assessment. The squared partial correlations showed that the keep-track variable represented a unique contribution of 10% of the variance, followed by numerical updating, which made a 9% contribution. Reasoning measures contributed an additional 5% of the variance (see Table 2). In sum, the most relevant predictors of teacher assessment were the updating variables.

## Discussion

The main objective of this study was to assess the relative capacity of two WMU tasks, the keep-track task and the numerical updating task, to the prediction of general academic attainment in typically developing children in a set of academic-related tasks. The results show that WMU accounts for an important percentage of unique variance in mathematical problem-solving performance, arithmetical operations and even in teacher assessment, thereby providing an additional and specific explanation for attainment. These results are in line with those reported in other studies involving typically developing children (e.g., St Clair-Thompson & Gathercole, 2006; Van der Sluis et al., 2007) and also to those obtained in studies of students with learning difficulties, in which WMU is highlighted as one of the key factors in academic attainment with respect to reading comprehension, mathematical problem-solving and arithmetic operations (e.g., Borella et al., 2010; Carretti et al., 2005; Cornoldi et al., 2012; Palladino et al., 2001; Passolunghi & Pazzaglia, 2004, 2005; Pelegrina, Capodieci, Carretti & Cornoldi, 2014; Swanson & Beebe-Frankenberger, 2004).

Our results indicate that both WMU tasks are predictive measures of academic attainment, although the numerical updating task was to some extent a more consistent predictor of children's performance. This may be due to the fact that this task requires content to be transformed in memory through numerical operations. Following the proposal of Ecker et al. (2010), access to memory contents, its transformation and, to a lesser degree, the substitution of obsolete information, affect the accuracy of WMU. Of these components, transformation has the greatest impact on updating accuracy and speed, which may mean that the numerical updating task accounted for unique variance in all of the academic measures.

Ecker et al., (2010) have argued that transformation might be related to general processing speed, which is one the main factors underlying individual differences. The predictive power of a task that requires transformation is therefore not surprising. In most academic tasks, the speed of modification of information maintained in memory may be critical to the final outcome. For instance, in order to understand a text, it is not only necessary to maintain relevant data and replace outdated information, but also to actively construct a dynamic representation of the content, which involves transforming the representation as the task progresses. Similarly, arithmetical operations, especially those involving various digits, require the transformation, maintenance and replacement of the initial numbers with intermediate values, and of these in turn with the final result.

Given that the different component processes of WMU might make distinct and independent contributions to performance, the role of transformation may be of relevance in an educational context. Moreover, it would be of interest to assess the extent to which the difficulty of the operation plays a role in the predictive power of the transformation component. The difficulty in carrying out the required transformation might determine the degree to which the task is sensitive to individual differences.

It is also noteworthy that there was no differential relationship between the WMU tasks and the academic-related measures in accordance with the type of information: numerical or verbal. Some studies which have included students with learning disabilities have shown that WM tasks are related to different degrees to complex tasks such as reading comprehension (e.g., Cain, Oakhill, & Bryant, 2004; Seigneuric, Ehrlich, Oakhill, & Yuill, 2000) and mathematical problem-solving (Hitch & McAuley, 1991; Iuculano, Moro, & Butterworth, 2011; Siegel & Ryan, 1989), depending specifically on whether the information is linguistic or numerical. However, other studies with children with difficulties (e.g. Passolunghi & Siegel, 2001; Turner & Engle, 1989) have not found this differential relationship (see also Peng & Fuchs, 2014). It is possible that individual differences in verbal and numerical tasks reflect differences in the ability to represent both types of information (Pelegrina et al., 2014). Conversely, when students do not have difficulties with such specific contents, domain-general mechanisms, in our case effectiveness in updating the information, would be the main determinants of task performance. It would be of interest to include the same WMU task in numerical and linguistic versions in order to better control the influence of the type of content to be updated on the performance of children both with and without specific difficulties.

This study also aimed to determine whether WMU accounted for unique variance in academic performance beyond that which could be accounted for by intelligence-related measures. It was found that WMU and intelligence shared a substantial percentage of the variance in academic attainment, which may indicate that both measures are influenced by certain common mechanisms which are in turn responsible for individual differences in cognitive performance.

Fluid intelligence and WMU could tap general aspects of cognitive control, where this may be the mechanism which explains its common predictive capacity for various indices of academic attainment (e.g., Steinmayr, Ziegler, & Träuble, 2010). It is a widely-held view that fluid cognition is strongly related to the executive functions of working memory (for reviews see Ackerman, Beier, Boyle, 2005, or Kane & Engle, 2002), although they make differential contributions to any explanation of academic attainment (e.g., Alloway & Alloway, 2010; Cain et al., 2004; Gathercole, Alloway, Willis, & Adams, 2006; Yuan, Steedle, Shavelson, Alonzo, & Oppezzo, 2006). In relation to WMU, it has been suggested that the same cognitive control ability underlies performance both in WM capacity tasks (e.g., complex span tasks) and in WMU tasks, and that this could be the reason why both types of tasks predict complex cognitive skills similarly well (Schmiedek et al., 2009). Therefore, it can be argued that fluid intelligence, which is associated with the ability to adapt thought flexibly to new situations or problems (Cattell, 1971), requires executive aspects of updating and cognitive control, independent of the content and specific domain of the task.

In addition to the shared variance with intelligence, WMU also made an independent contribution to academic achievement prediction. Thus, some cognitive operations specifically involved in updating appear to be of especial relevance for performance in academic tasks. This suggests that WMU tasks may be useful as assessment tools in educational screening tests, both in predicting learning attainment and as a way of identifying children with possible learning difficulties.

This study has some limitations that should be addressed by future research. The specific contribution of each updating component to the prediction of academic attainment could be assessed with the same task. This would allow for better isolation of the contribution of each component of WMU, and identification of its role in determining general and stable individual differences. Alternatively, several concurrent measures of these components in different tasks could be considered.

Another issue to explore concerns whether there are age-related differences in the specific contribution of the different updating tasks. In this study only 9-year-old children participated. It could be relevant to assess children of different ages, since previous research has shown that children of different ages may rely on different WM subcomponents (e.g., Andersson & Lyxell, 2007; Kyttäla, Aunio, & Hautamäki, 2010). In fact, the relevance of different types of information appears to change according to age. For instance, younger children rely more on visual representations, but as they get older they tend to use verbal strategies (e.g., rehearsal) that allow them to recode visual information in verbal codes (Alloway, Gathercole, & Pickering, 2006; Rasmussen & Bisanz, 2005). Moreover, the predictive relationship between verbal and non-verbal WM abilities and academic attainment appears to show a degree of domain specificity depending on age (Jarvis & Gathercole, 2003). Therefore, future research could analyze updating performance with different types of information, in children of different ages, and its relationship to academic achievement in different domains.

We can conclude that performance in WMU tasks accounts for an important proportion of the variance in the performance of mathematical problems, arithmetical operations, verbal comprehension, and even in overall teacher assessment. In many of the tasks which are carried out in the school, it is necessary to access or retrieve the content of memory, to manipulate or transform this content, and to replace or substitute information which is no longer relevant to the task. Therefore, a low ability to update quickly and effectively the content of working memory can act as a bottleneck, making it difficult for students to carry out their schoolwork effectively.

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**Notes**

[1] In both WMU tasks the number of items to be recalled was manipulated. In both WMU tasks, participants performed better in the low load versus the high load condition, which shows that the difficulty manipulation was effective: 60.18 compared with 46.57 for the keep-track task, $F(1, 96) = 115.147$; $Mce = 78$; $p < .001$; $\eta 2 = .545$, and 56.88 compared with 42.19 for the numerical updating task, $F(1, 96) = 135.676$; $Mce = 77.058$; $p < .001$, $\eta 2 = .841$. Means obtained for each task indicate that all of the conditions, even those considered low load, entail a considerable level of difficulty. However, the analysis of the manipulation effect of the load did not provide relevant additional information in terms of the questions discussed here, and so this factor was not included in the analyses.

Figure 1

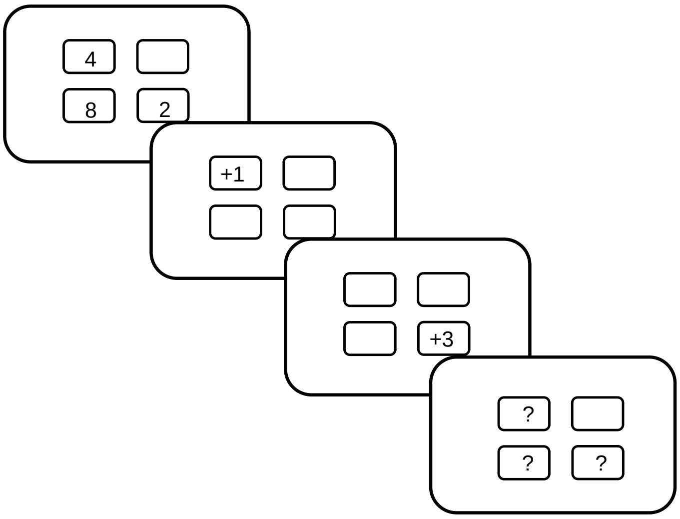


Figure 1. Example of one trial in the numerical updating task.

**Table captions**

Table 1. Means, standard deviations and bivariate correlations for academic attainment, WMU tasks and intelligence measures.

Table 2. Regression analyses predicting academic attainment by WMU tasks and intelligence measures.

Table 1

| *Measures* | *M* | *SD* | 2 | 3 | 4 | 5 | 6 | *7* | *8* |
|---|---|---|---|---|---|---|---|---|---|
| *Academic attainment* | | | | | | | | | |
| 1. Mathematical problems | 16.01 | 5.59 | .67** | .58** | .59** | .57** | .36** | .52** | .63** |
| 2. Arithmetic operations | 18.57 | 3.94 | - | .51** | .55** | .31** | .44** | .51** | .48** |
| 3. Verbal comprehension | 15.54 | 4.15 | | - | .48** | .32** | .53** | .34** | .45** |
| 4. Teacher assessment | 6.74 | 2.02 | | | - | .53** | .37** | .57** | .61** |
| *Intelligence* | | | | | | | | | |
| 5. Reasoning | 18.68 | 4.25 | | | | - | .30** | .46** | .56** |
| 6. Vocabulary | 8.23 | 3.02 | | | | | - | .35** | .40** |
| *Updating* | | | | | | | | | |
| 7. Keeping track | 53.38 | 13.93 | | | | | | - | .52** |
| 8. Numerical updating | 49.54 | 21.61 | | | | | | | - |

Note. ** $p < .001$.

Table 2

| | | $R^2$ | $B$ | $t$ | $pr^2$† |
|---|---|---|---|---|---|
| *Mathematical Problems* | | .50 | | | |
| | Numerical updating | | .36 | 3.70** | ,13 |
| | Keeping track | | .18 | 2.00* | ,04 |
| | Reasoning | | .26 | 2.86** | ,08 |
| | Vocabulary | | .08 | .93 | ,01 |
| *Arithmetical operations* | | .37 | | | |
| | Numerical updating | | .24 | 2,14* | ,05 |
| | Keeping track | | .32 | 3,20** | ,10 |
| | Reasoning | | -.04 | -,43 | 0 |
| | Vocabulary | | .25 | 2,68** | ,07 |
| *Verbal comprehension* | | .35 | | | |
| | Numerical updating | | .23 | 2,08* | ,04 |
| | Keeping track | | .05 | ,53 | 0 |
| | Reasoning | | .05 | ,45 | 0 |
| | Vocabulary | | .40 | 4,26** | ,16 |
| *Teacher assessment* | | .49 | | | |
| | Numerical updating | | .31 | 3,10** | ,09 |
| | Keeping track | | .29 | 3,22** | ,10 |
| | Reasoning | | .20 | 2,16* | ,05 |
| | Vocabulary | | .08 | 1,02 | ,01 |

*Note*. **$p<.001$; *$p<.05$. † $pr^2$ = squared part correlations, represents the unique contribution for each variable.